

Efficient Attention Mechanisms for Long-Context Language Models: A Survey and Benchmark

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Abstract

We present a comprehensive survey and benchmark of efficient attention mechanisms for long-context language models. Our analysis covers linear attention, sparse attention, and hybrid approaches across multiple domains including natural language understanding, code generation, and multi-document summarization. We evaluate 15 methods on standardized benchmarks and provide practical guidelines for practitioners.

1. Introduction

The rapid growth of large language models (LLMs) has created an urgent need for efficient attention mechanisms that can handle increasingly long context windows. Standard quadratic attention becomes prohibitively expensive beyond 100K tokens...